

Waveform analysis with PSpice

Waveform analysis with PSpice

What is waveform analysis?

After the simulation is complete, PSpice plots the resulting waveforms so you can visualize the behavior of the circuit and determine the validity of your design.

Render the results of the meta-simulation This means you can plot additional information derived from the waveforms. What you can design depends on the types of analyzes you are dealing with. Bode plots , phase margins, derivatives for microsignal characteristics, waveform families, and histograms are just some of the possibilities. You can also plot other waveform characteristics such as rise time vs. temperature, or exact overshoot vs. element value.

Identify Design Errors in Digital Circuits When PSpice detects setup and hold violations, flow states, or timing hazards, a detailed message is displayed along with the corresponding waveform. PSpice also helps you identify the problem in your design .

Chapter 1 Things you need to know

Using PSpice with Orcad programs

Use Capture to prepare the simulation

Capture is a design input program that you need to prepare your circuit for simulation. This means:

- placement and connection of component symbols,
- setting element values and other attributes;
- defining the imported waveforms,
- enable one or more analyses, and
- note the circuit points you want to see results on.

Capture is also the control point for running other programs used in the simulation design flow.

What is PSpice Stimulus Editor?

The Stimulus Editor is a graphical waveform input editor that allows you to specify the shape of the time-dependent signals used to examine the response of your circuit during simulation. Using the Trigger Editor, you can specify:

- analog stimuli with sine wave, pulse, line segment, exponential pulse, single frequency FM patterns, and
- digital stimuli ranging from simple clocks to complex pulse patterns and bus sequences.

The Stimulus Editor allows you to plot analog line segment and all digital stimuli by clicking on the points along with the timeline that corresponds to the input values you want to transmit.

Files needed for the simulation

What is PSpice Model Editor?

The PSpice Model Editor is a model exporter that creates model definitions for PSpice A/W use during simulation. What the PSpice Model Editor needs is information about the device found in the standard datasheets. As you enter datasheet information, the Model Editor displays device characteristic curves so you can confirm device behavior on a model-by-model basis. When you're done, the PSpice Model Editor automatically creates a part for the model so you can immediately use the model-based part in your design.

Useful files for the simulation

To simulate your design, PSpice needs to know:

- the pieces of your circuit and how they connect,
- what analyzes to perform,
- the simulation models that correspond to the parts of your circuit, and
- the definitions of the stimuli with which you will test.

This information is accessible from various data files. Some of these are generated by Capture , others come from libraries (which can also be generated by other programs, such as PSpice Stimulus Editor and PSpice Model Editor), and others are user-defined.

Files that Capture creates

When you start the simulation process, Capture first creates files that describe parts and connections within your circuit. These files are the netlist file and the circuit file that PSpice reads before doing anything else.

Athanasίου Vasileios Evangelos

19390005

Pages 49 - 51